

PHYSM0071: Coursework 1
Correlations and fluctuations in a two-dimensional lattice gas
Model report for marking (base seed 2468135)

Mapping and predictions

Let $V = L^2$, let the square-lattice coordination number be $q = 4$, and write $c_i = (1 + s_i)/2$. Since there are $qV/2$ bonds,

$$\sum_{\langle ij \rangle} c_i c_j = \frac{1}{4} \sum_{\langle ij \rangle} s_i s_j + \frac{q}{4} \sum_i s_i + \frac{qV}{8}, \quad N = \frac{V}{2} + \frac{1}{2} \sum_i s_i.$$

The grand-canonical Hamiltonian therefore becomes

$$E - \mu N = -\frac{\epsilon}{4} \sum_{\langle ij \rangle} s_i s_j - \left(\frac{q\epsilon}{4} + \frac{\mu}{2} \right) \sum_i s_i - \frac{qV\epsilon}{8} - \frac{\mu V}{2}.$$

Comparison with $\mathcal{H}_I = -J \sum_{\langle ij \rangle} s_i s_j - H \sum_i s_i$ gives $J = \epsilon/4$, $H = q\epsilon/4 + \mu/2$, and the configuration-independent term $C = -qV\epsilon/8 - \mu V/2$. The factor $\exp(-\beta C)$ cancels from normalised probabilities, so C does not affect configurational averages or phase behaviour, although it is required when relating the two partition functions. Coexistence corresponds to Ising zero field, so $\mu_{\text{coex}} = -q\epsilon/2 = -2\epsilon$. Also

$$\rho = \frac{1+m}{2}, \quad k_B T_c = \frac{2J}{\ln(1+\sqrt{2})} = \frac{\epsilon}{2 \ln(1+\sqrt{2})} = 0.567296 \epsilon.$$

Mean-field theory gives $m = \tanh[\beta(qJm + H)]$, hence $k_B T_c^{\text{MF}} = qJ = \epsilon$. It overestimates T_c because it replaces the fluctuating local environment by its mean and neglects spatial correlations.

For a trial flip, put $\delta = 1 - 2c_i = \Delta N$ and let n_i be the number of occupied neighbours. Then

$$\Delta E = -\epsilon \delta n_i, \quad \Delta(E - \mu N) = -\delta(\epsilon n_i + \mu).$$

These are respectively `delta_E`, `delta_N` and `delta_K` in `local_grand_energy_change`.

Validation

After 2000 equilibration sweeps, 12,000 measurements separated by five sweeps gave:

μ	$\langle N \rangle$, simulation	exact	$\text{var}(N)$, simulation	exact
-1	107.522(85)	107.577	78.83(99)	78.64
0	200.056(91)	200.000	99.43(1.28)	100.00
1	292.360(80)	292.423	77.49(91)	78.64

Uncertainties are standard errors from block means (or block variances), using 20-measurement blocks. The stationary trace and full $P(N)$ at $\mu = -1$ agree with the binomial law (Fig. 1).

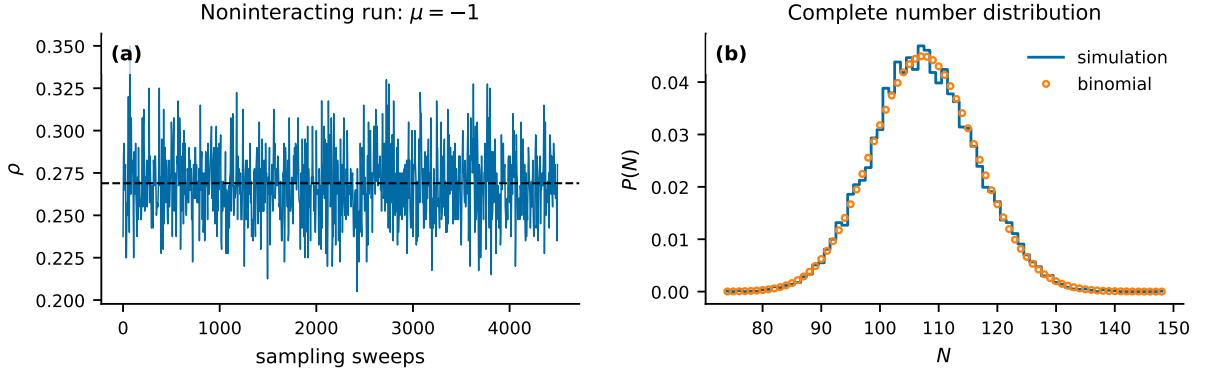


Figure 1: **Validation.** (a) Representative noninteracting density trace after equilibration. (b) The complete measured distribution at $\mu = -1$ and the exact binomial result.

Equation of state and density response

The two equations of state are antisymmetric about $(\mu_{\text{coex}}, \rho) = (-2, 1/2)$ (Fig. 2a). Paired densities satisfy $\rho(-2 + \Delta\mu) + \rho(-2 - \Delta\mu) = 1$ within uncertainty: the largest standardised deviation is 2.0, consistent with the expected scatter among several comparisons. The response becomes much steeper on approaching T_c .

At coexistence the three response estimates were:

T/T_c	finite difference	$\beta V \text{var}(\rho)$	$\beta S(0)$
1.5	1.994(55), $h = 0.01$	2.002(30)	2.002(30)
1.15	13.06(52), $h = 0.005$	13.44(37)	13.44(37)

At $1.5T_c$, a central difference is stable for $h = 0.01$ – 0.02 and agrees with the fluctuation result. At $1.15T_c$, the estimates for $h = 0.05, 0.02, 0.01, 0.005$ are 6.43, 10.25, 11.77, 13.06. Thus a broad equation-of-state grid still underestimates the derivative, but the finest estimate agrees with the fluctuation result within the combined uncertainty. The variance and $S(0)$ estimates coincide because, with the program's normalization, $S(0) = \text{var}(N)/V = V \text{var}(\rho)$; this is an exact normalization check rather than an independent measurement.

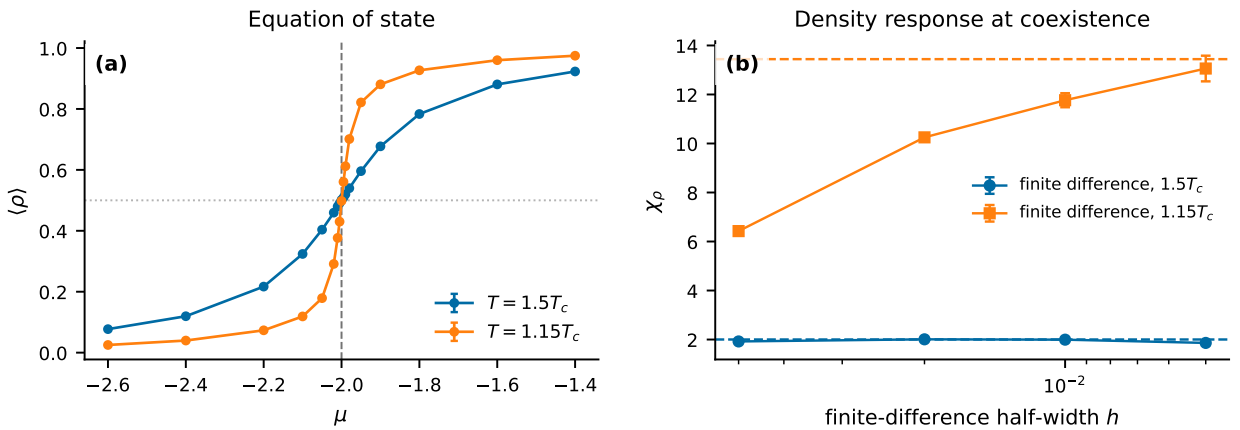


Figure 2: **Equation of state and response.** (a) Density against chemical potential; error bars are mostly smaller than the symbols. (b) Convergence of the symmetric finite-difference derivative as its half-width h is reduced. Dashed horizontal lines are the fluctuation estimates.

Spatial correlations

The configurations develop progressively larger domains as T_c is approached (Fig. 3a–d), accompanied by a pronounced enhancement of $S(k)$ at small k . The returned two-dimensional connected structure factor was averaged over exact square-lattice shells,

$$k = \frac{2\pi}{L} \sqrt{n_x^2 + n_y^2}.$$

Six independent runs were averaged at each temperature. Fits to $S(k) = A/[1 + (k\xi)^2]$ used the first six shells, $0 \leq k \leq 0.7405$; uncertainties were obtained by fitting the independent runs separately:

T/T_c	ξ	bootstrap s.e.
1.50	1.459	0.010
1.25	2.606	0.012
1.10	5.801	0.073
1.05	9.968	0.058

At $1.05T_c$, changing the upper fit limit from $k_{\max} = 0.55$ to 0.95 gives $\xi = 9.792$ – 10.108 . This sensitivity, together with ξ becoming comparable with $L/2$, indicates finite-size effects and deviations from the simple small- k fitting form. A log–log fit gives an effective $\nu_{\text{eff}} \simeq 0.84$, but finite-size effects, corrections to scaling and the short temperature interval dominate. The growth is stronger than the mean-field $t^{-1/2}$ prediction and qualitatively closer to the Ising t^{-1} form, but the data do not justify a precise exponent.

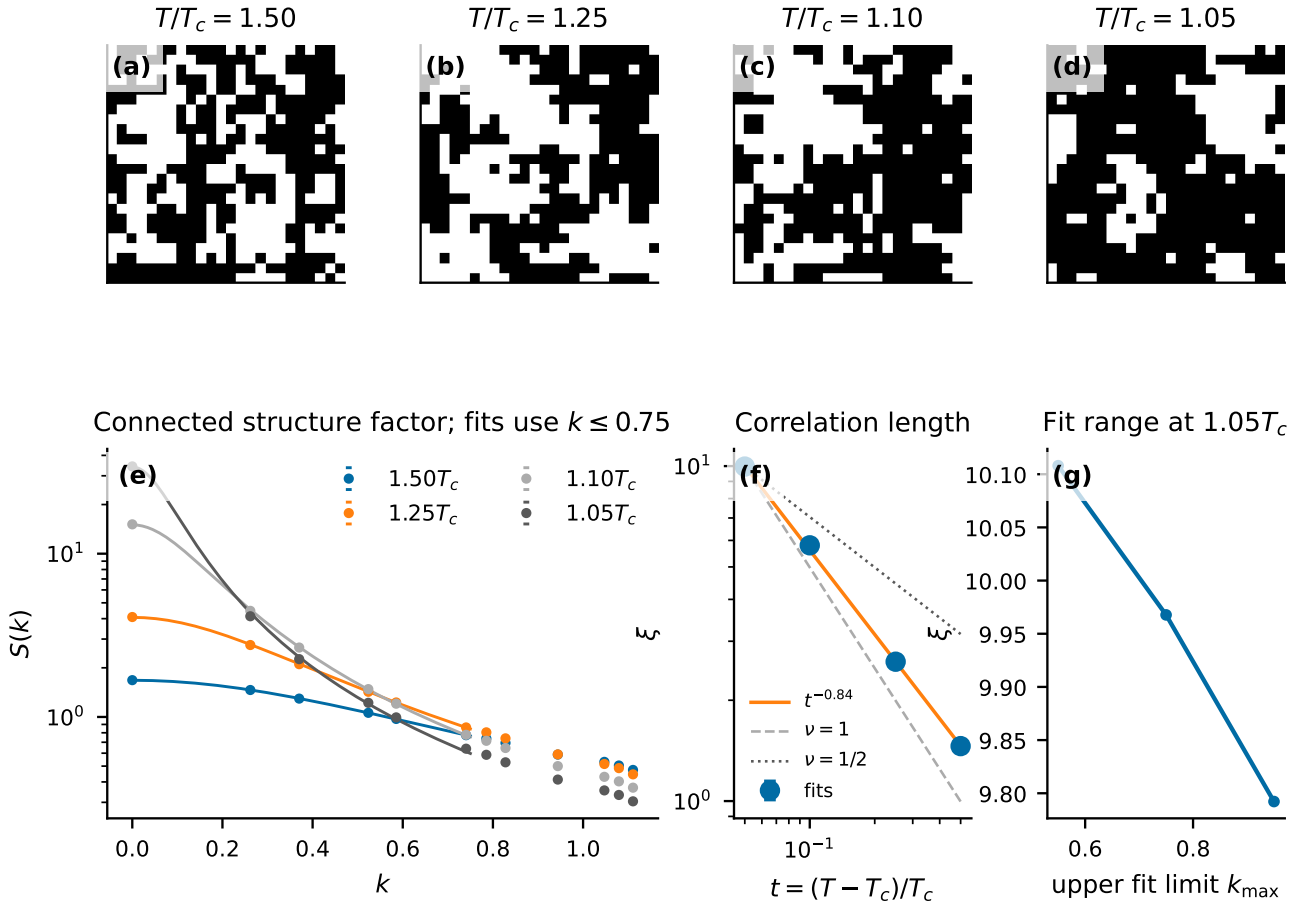


Figure 3: **Spatial correlations.** (a–d) Representative equilibrated configurations. (e) Radially averaged connected structure factors and the fitted low- k forms. (f) Extracted correlation lengths with exact and mean-field guide slopes. (g) Sensitivity to the upper fit limit at $1.05T_c$.

Density distributions and finite-size effects

At $1.2T_c$, $P_L(\rho)$ is a single broad peak centred at $1/2$. At T_c , the distribution is broad and double peaked. Below T_c , the two sector-conditioned peaks are narrow and lie near $\rho \simeq 0.05$ and 0.95 . The low-start run changed sector three times, whereas the high-start run did not. After applying $\rho \rightarrow 1 - \rho$ to the high-density distribution, its peak agrees with the low-density distribution within the sampling noise, as required by particle-hole symmetry. The residence times themselves are not reliable estimates of the equilibrium sector weights.

An equilibrium average $\langle \rho \rangle = 1/2$ below T_c would consequently describe an equal mixture of low- and high-density states, not typical homogeneous configurations at density $1/2$. Individual configurations reside overwhelmingly in one phase; the mean lies between the peaks because it averages over phase identity.

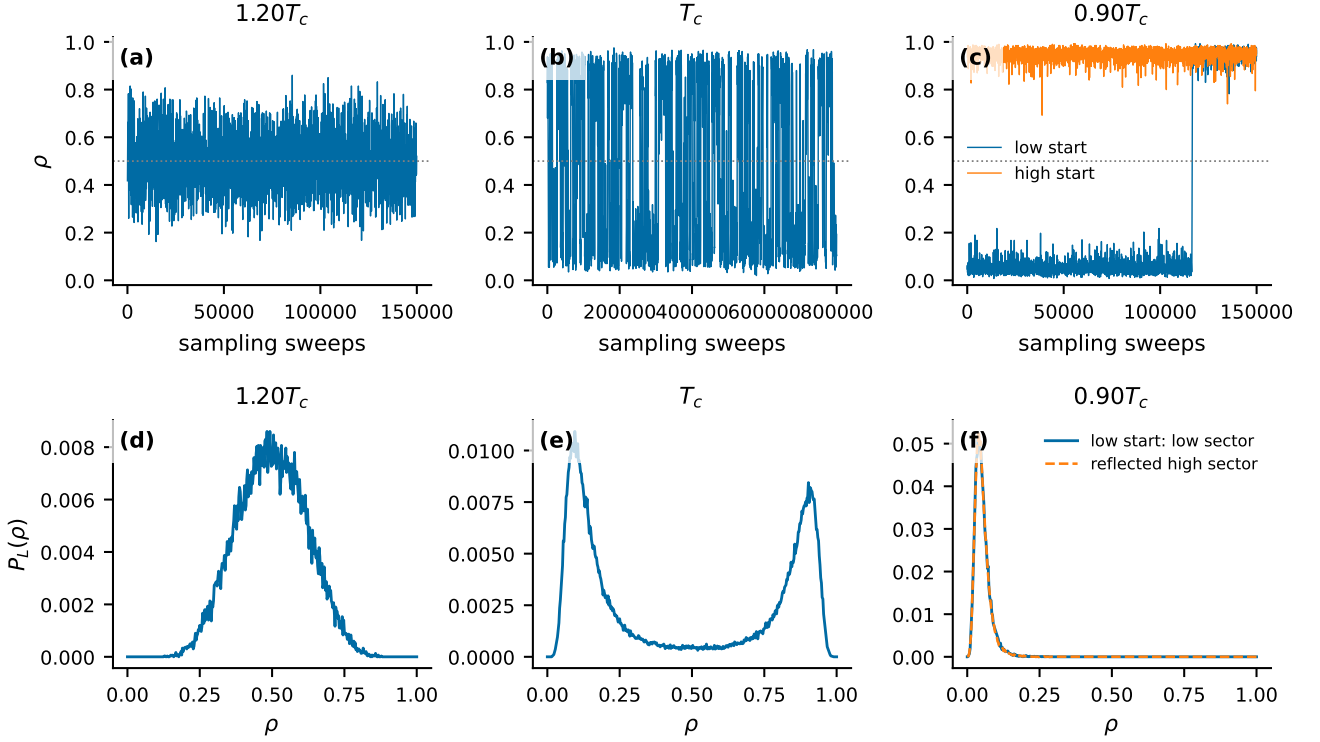


Figure 4: **Evolution of the density distribution.** (a–c) Density histories at coexistence. The subcritical low-start run changes sector while the high-start run remains trapped. (d,e) Measured distributions above and at T_c . (f) The two subcritical sector-conditioned distributions, compared after reflecting the high-density result.

At criticality, the peaks of $P_L(\rho)$ move towards $1/2$ and the distribution narrows as L increases (Fig. 5a). This model calculation averaged two independent 40,000-measurement runs for each size, more sampling than the minimum specified in the exercise:

L	U_4	block-bootstrap s.e.
12	0.61064	0.00114
16	0.61131	0.00144
20	0.61110	0.00171
24	0.61110	0.00218

All values agree with $U_4^* = 0.61069$; any finite-size drift is smaller than the present sampling uncertainty. Since U_4 measures dimensionless distribution shape, its approximate independence of L is evidence of scale invariance. In real space this means that critical configurations have no characteristic domain size: density fluctuations occur over a broad range of scales, up to the system size. The uncertainty grows with L because the dynamics slows. For the extended $L = 24$ diagnostic run, the block-bootstrap uncertainty became stable as the block length increased. Both signs of m were sampled, with 1355 crossings, and the first- and second-half distributions differ by only 0.022 in total variation. Their mean densities, 0.440 and 0.449, also agree within the slow fluctuations. These checks support the reported even moments and distribution shape, although the relative sector weights remain less well determined.

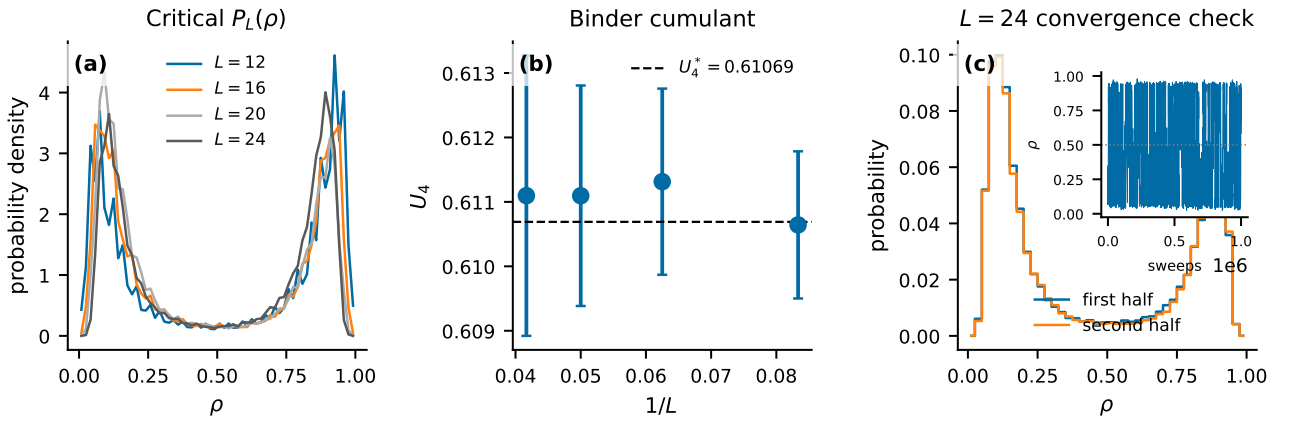


Figure 5: **Critical finite-size behaviour.** (a) Order-parameter distributions. (b) Binder cumulants and the periodic-square reference. (c) First/second-half comparison for the extended $L = 24$ run; inset: its density history.

Production-run record. All seeds are $s = (2468135 + 104729r) \bmod (2^{32} - 1)$, where r is the run ID. All interacting runs use $\epsilon = 1$, and ρ_0 denotes the initial density. The detailed individual record is supplied in `production_run_table.csv`; the complete grouped record used above is:

Purpose; run IDs	$L, T, \mu; \rho_0$	equilibration; interval; measurements	seed range; acceptance
ideal validation; 1–3	$20, 1, \{-1, 0, 1\}; \epsilon = 0, \rho_0 = p$	2000; 5; 12000	2572864–2782322; 0.538–1.000
equation of state; 10–31	$24, \{1.5, 1.15\}T_c, -2 + \Delta\mu; \rho_0 = .15/.5/.85$	6000; 5; 12000	3515425–5714734; 0.050–0.533
fine response; 40–51	as above, $\Delta\mu = \pm.02, \pm.01, \pm.005$	6000–8000; 5; 12000–20000	6657295–7809314; 0.324–0.532
correlations; 100–123	$24, T/T_c = 1.50, 1.25, 1.10, 1.05; \mu = -2, \rho_0 = .5$; six replicas	8000; 5; 10000	12941035–15349802; 0.262–0.534
distribution evolution; 200, 202–203	$20, \{1.2, 0.9\}T_c, -2; \rho_0 = .5/.1/.9$	6000–10000; 5; 30000	23413935–23728122; 0.090–0.394
critical sizes; 300–307	$L = 12, 16, 20, 24; T_c, -2; \rho_0 = .5$; two replicas	12000; 5; 40000	33886835–34619938; 0.181–0.191
extended critical; 320–321	$L = 20, 24; T_c, -2; \rho_0 = .5$	12000; 5; 160000/200000	35981415–36086144; 0.189–0.190

Errors for ordinary means and variances were obtained from stable block estimates. Correlation-length errors resample the six independent runs; U_4 errors use a blocked bootstrap of the available trajectories. Raw trajectories were retained for every entry.

Marking-template note: this model solution was generated for use in marking and therefore does not reproduce the student’s required signed AI-use declaration.